SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing

SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

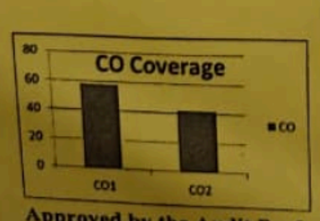
Academic Year: 2025-26 (ODD)

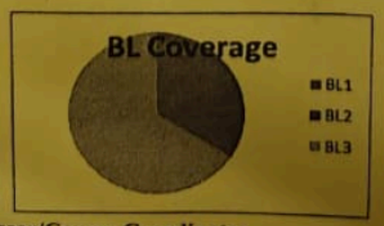
Set C

Test: CLA-T1 | **Date: 11-08-2021**
Course Code & Title: 21CSS201T/Computer Organization and Architecture | **Duration:** 2 Hours
Year & Sem: II Year / III Sem | **Max. Marks:** 50

Part – B
Answer any FOUR Questions
(4x 10 = 10 Marks)

Q. No	Question	Marks	BL	CO	PO	PI Code
6	i) Subtract –57 from 12 using two's complement notation. (3 marks) ii) Find the two's complement of the negative number (−57). (4 marks) iii)Two unsigned 8-bit binary values are stored as 00001100 and 00111001. If these values are added, what will be the resulting binary word and its decimal equivalent? (3 marks)	10	3	1	2	1.1.1
7	A calculator internally represents decimal digits in BCD. If it is given the inputs 162.27 and 349.32, demonstrate the sequence of operations it performs to display the correct sum. (10 marks)	10	3	1	2	1.1.1
8	i)A certain binary-coded value is stored as 11110110. If the system decides to re-encode it using Excess-3 representation, what would be the resulting code word? (4 marks) ii)A processor receives two operands, 11001 and 11011, both in binary. If the arithmetic unit multiplies them directly in binary form, what product will be generated? (3 marks) iii)A floating value is expressed in hexadecimal as $(DF2.53E)_{16}$. Determine its equivalent numerical value in the decimal system. (3 marks)	10	3	1	2	1.1.2
9	Discuss how an instruction fetch and execute takes place for an instruction ADD R1, LOCA.	10	2	2	1	2.1.2
10	Describe the different types of addressing modes used in computer architecture. Give suitable examples for each.	10	2	2	1	2.1.2





Approved by the Audit Professor/Course Coordinator

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